# Muhammad Talha Atif

Full Stack Engineer

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## SUMMARY

Full Stack Engineer with 1+ year of production experience building and shipping end-to-end web systems across backend services, frontend interfaces, and cloud infrastructure. Delivered distributed microservices and REST APIs for an enterprise Identity Governance platform, and shipped a complete full stack social platform serving real users daily. Comfortable owning the full request lifecycle from database schema to UI layer, with a consistent focus on system reliability, clean architecture, and production-quality engineering across every layer of the stack.

## WORK EXPERIENCE

**Full Stack Engineer** *Dec '25 to Present*
**MicroTechX** Remote

*Built and maintained backend services and frontend modules for an enterprise IAM platform, owning distributed microservices, event-driven workflows, REST APIs, and a React/Next.js dashboard across a compliance-heavy Identity Governance system.*

- Architected Kafka-based event-driven microservices for IAM audit trails and approval workflows across 8+ distributed services, configuring consumer concurrency, retry windows, and dead-letter queues for fault-tolerant async processing.
- Designed a bulk role assignment API supporting 5,000+ users in a single operation using batch queries, and redesigned the IAM web dashboard with standardised design tokens and a unified component architecture, eliminating visual inconsistency across the entire platform.
- Led a zero-downtime OpenSearch migration reindexing approximately 5M+ documents using index templates and alias switching, and migrated high-cardinality fields from text+keyword to keyword for improved query performance across the analytics layer.

**Full Stack Engineer** *Sep '25 to Jan '26*
**Digital Upgraders LLC** Remote

*Built and maintained backend APIs and frontend features for a production social platform serving photographers, videographers, and creative professionals across web and mobile.*

- Resolved a feed API performance bottleneck by identifying missing indexes on read-heavy columns, removing unused fields from response DTOs, and adding a 5-minute in-memory post cache, reducing response time from 3 seconds to 700ms and directly improving the frontend feed experience.
- Built Reported Posts and Comments systems end to end, database schema, REST APIs with ranking logic, and admin moderation UI, while enforcing duplicate-report prevention and accurate count validation across all edge cases.
- Led zero-downtime production migration to Fly.io covering Docker configuration, DNS setup, Supabase production environment, email provider migration, and environment management across a live multi-service platform.

## PROJECTS

**Moodly** | AI Experience Booking Platform | Demo

- Designed a multimodal emotion detection pipeline processing voice recordings and facial images through separate classification models, routing results via an async message queue to a booking recommendation engine that selects experiences based on detected mood state.
- Built RESTful backend APIs for the full booking lifecycle and a cross-platform React Native frontend with one-tap booking flows, mood visualisation components, and a voice-command interface for hosts, all connected through a real-time event-driven architecture.
- Implemented strict input validation, secure authentication, and unit-tested business logic across all backend critical paths, while enforcing TypeScript-strict component architecture on the frontend to reduce edge-case bugs in production.

**Memingo** | Language Learning Platform | Demo

- Identified and resolved a critical leaderboard bottleneck where rankings were recalculated on every read request,

replaced with a write-time precomputed cache, reducing API response time from 30 seconds to 90ms and eliminating the drop-off on the rankings screen.

- Built 9 interactive lesson format components on the frontend and a real-time object detection module on the backend that intercepts camera frames, runs item classification, and returns vocabulary and cultural context as a single structured API response.
- Integrated a conversational AI chatbot with session-aware context management and structured output parsing, delivering personalised learning recommendations that adapt based on the learner's progression state across sessions.

## SKILLS

**Backend** Java, Spring Boot, Spring MVC, Spring Security, Node.js, Express.js, NestJS, FastAPI, Python, REST APIs, gRPC

**Frontend** React.js, Next.js, React Native, TypeScript, JavaScript, Tailwind CSS, Shadcn UI, Framer Motion

**Architecture** Microservices, Event-Driven Architecture, Distributed Systems, Kafka, RabbitMQ, WebSockets

**Databases** PostgreSQL, MySQL, MongoDB, SQLite, Redis, Supabase, Firebase, Hibernate/JPA, OpenSearch

**Security** JWT, OAuth 2.0, RBAC, Rate Limiting, Spring Security, Input Validation

**Cloud** AWS (S3, RDS), Firebase, Supabase, Dokploy, Fly.io

**DevOps** Docker, GitHub Actions, CI/CD, Maven, Git

**Tools** Postman, Swagger, Stripe, VS Code, IntelliJ IDEA

## AWARDS

**Runner-up, App Development Hackathon (2025)** *May '25*
Organized by Punjab University

**Ranked, ICPC Asia Topi Online Programming Contest (2024)** *Nov '24*
Organized by International Collegiate Programming Contests

## EDUCATION

**Bachelor's degree in Computer Science, University of Management and Technology – UMT** *Oct '21 to Aug '25*